Influence of keeping pheasants in captivity vs. nature on the biological value of meat and its use in human nutrition.

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The life of game birds (pheasants) in nature is coupled with a number of difficulties in all seasons of the year. This refers to finding food, breeding, laying eggs, raising the young, fleeing from their natural enemies and lack of protection from unfavorable climatic conditions. The pheasants that live in captivity--aviaries for pheasants--do not have such difficulties--they are fed regularly by quality feed for pheasants, they are protected from bad weather and natural enemies. Our research was aimed at determining the biological value of meat of pheasants grown in the two different settings--in captivity and in nature. The highest weight achieved wild pheasant males (1232.4 +/- 147.36 g). The differences between tested pheasant groups were statistically very high significant ($P < 0.001$). The differences between groups related to breast weight and tights with drumsticks weight were statistically very high significant ($P < 0.001$). Between breast parts (%) and legs parts (%) were notified very high ($P < 0.001$) i.e. high ($P = 0.002$) differences. The highest weight breast muscles and tights with drumsticks had wild pheasants (282.6 +/- 63.53 g i.e. 206.2 +/- 37.88g). Wilde pheasants had lower part (%) and lighter (g) skin with subcutaneous fatty tissue on breasts. Female pheasants cultivated on both ways had higher skin part (%) and subcutaneous fatty tissue in tights with drumsticks. Related to chemical composition of breast muscles is established statistically significant differences ($P < 0.001$ i.s. $P = 0.040$)) in part of Ca (%) and P (%). In wild pheasant tights with drumsticks muscles established statistically very significant ($P < 0.001$) higher part of moisture, protein and Ca, i.e. statistically very high significant ($P < 0.001$) lower part of fat and energetic value. Research results indicate that the quality of meat of pheasants grown in nature has higher biological value than the meat of pheasants kept in aviaries, which means it has advantages in human nutrition.